

Appendix A: Material Properties

Appendix A: Material Properties Reference

A.1 Structural Metals

A.1.1 Ferrous Metals (Iron-Based)

Material	Density (kg/m ³)	Young's Modulus (GPa)	Yield Strength (MPa)	Tensile Strength (MPa)	Thermal Expansion (x10 ⁻⁶ /°C)	Thermal Conductivity (W/(m*K))
Mild Steel (A36)	7,850	200	250	400-550	11.7	50
Structural Steel (A572 Gr50)	7,850	200	345	450	11.7	50
4140 Steel (Annealed)	7,850	200	415	655	11.5	42
4140 Steel (Q&T)	7,850	200	1,000-1,400	1,100-1,600	11.5	42
1018 Cold Rolled	7,870	205	370	440	11.7	52
Stainless 304	8,000	193	215	505	17.3	16
Stainless 316	8,000	193	205	515	16.0	16
Stainless 17-4 PH (H900)	7,800	197	1,170	1,310	10.8	17
Cast Iron (Gray, Class 30)	7,200	100-140	200-300	210-340	10.8	50

Material	Density (kg/m ³)	Young's Modulus (GPa)	Yield Strength (MPa)	Tensile Strength (MPa)	Thermal Expansion (x10 ⁻⁶ /°C)	Thermal Conductivity (W/(m*K))
Cast Iron (Ductile, 60-40-18)	7,100	169	276	414	11.8	36
Tool Steel (D2, Hardened)	7,700	210	1,700-2,000	1,900-2,200	11.5	20

Notes: - **Mild Steel (A36):** Most economical structural material. Good weldability. Used for frames, gantry beams, base plates. - **A572 Grade 50:** Higher strength than A36 (50 ksi vs. 36 ksi). Better strength-to-weight ratio. Common in structural tubing. - **4140 Alloy Steel:** Medium-carbon chromium-molybdenum alloy. Excellent toughness when heat-treated. Used for shafts, spindles, high-stress components. - **Stainless 304/316:** Austenitic stainless. Corrosion-resistant. Non-magnetic. Work-hardens during machining (difficult to cut). Used in food/medical/marine applications. - **17-4 PH:** Precipitation-hardening stainless. Combines corrosion resistance with high strength. Used for shafts, fasteners in corrosive environments. - **Gray Cast Iron:** Excellent vibration damping (10x steel). Low cost. Used for machine beds, bases. Not weldable. - **Ductile Iron:** Better ductility than gray iron. Weldable. Used for gears, couplings, heavy-duty frames.

A.1.2 Non-Ferrous Metals (Aluminum, Copper, Titanium)

Material	Density (kg/m ³)	Young's Modulus (GPa)	Yield Strength (MPa)	Tensile Strength (MPa)	Thermal Expansion (x10 ⁻⁶ /°C)	Thermal Conductivity (W/(m*K))
Aluminum 6061-T6	2,700	69	276	310	23.6	167
Aluminum 7075-T6	2,810	72	503	572	23.4	130
Aluminum 2024-T3	2,780	73	345	483	22.9	121
Aluminum 5083-H111	2,650	70	145	290	23.8	117
Aluminum Plate Tooling (MIC-6)	2,700	69	172	228	23.6	155

Material	Density (kg/m ³)	Young's Modulus (GPa)	Yield Strength (MPa)	Tensile Strength (MPa)	Thermal Expansion (x10 ⁻⁶ /°C)	Thermal Conductivity (W/(m*K))
Copper C110 (Pure)	8,940	120	70	220	16.5	391
Brass C360 (Free-Machining)	8,500	97	125	340	20.5	115
Bronze C932 (Bear-ing Bronze)	8,800	103	172	379	18.0	59
Titanium Ti-6Al-4V (Annealed)	4,430	114	880	950	8.6	7.2

Notes: - **6061-T6:** General-purpose structural aluminum. Good corrosion resistance, weldable. Used for gantry beams, plates, brackets. 1/3 weight of steel. - **7075-T6:** Aircraft-grade aluminum. Highest strength aluminum alloy. Poor weldability. Used for high-stress components where weight matters. - **2024-T3:** High-strength aluminum with copper. Excellent fatigue resistance. Used in aircraft structures. Not very corrosion-resistant (requires coating). - **5083:** Marine-grade aluminum. Best corrosion resistance. Non-heat-treatable. Used for waterjet tanks, enclosures in corrosive environments. - **MIC-6 (Cast Aluminum Plate):** Stress-relieved casting. Excellent flatness and stability. Used for machine beds, fixture plates, optical tables. - **Titanium Ti-6Al-4V:** Aerospace grade. High strength-to-weight ratio. Excellent corrosion resistance. Difficult to machine (low thermal conductivity causes tool heating). Used in high-performance applications.

A.1.3 Comparison Chart: Strength-to-Weight Ratio

Specific Strength = Yield Strength / Density (higher is better for weight-critical applications)

Material	Yield Strength (MPa)	Density (kg/m ³)	Specific Strength (kN*m/kg)
Aluminum 7075-T6	503	2,810	179
Titanium Ti-6Al-4V	880	4,430	199
4140 Steel (Q&T)	1,200	7,850	153
Aluminum 6061-T6	276	2,700	102
Mild Steel (A36)	250	7,850	32

Conclusion: Titanium and 7075 aluminum offer best specific strength, but titanium is 10-20x more expensive and difficult to machine.

A.2 Engineering Plastics and Composites

A.2.1 Thermoplastics

Material	Density (kg/m ³)	Young's Modulus (GPa)	Tensile Strength (MPa)	Max Temp (°C)	Water Absorption (%)	Applications
UHMW Polyethylene	930	0.8	22	80	<0.01	Low-friction bearing surfaces, chip deflectors, wear strips
Acetal (Delrin)	1,420	3.1	70	90	0.25	Gears, bushings, low-load bearings, insulators
Nylon 6 (PA6)	1,140	2.8	80	120	1.5-2.5	Cable carriers, wear strips, gears (requires lubrication)
Nylon 6/6 (PA66)	1,150	2.9	83	130	1.5-2.5	Higher strength than Nylon 6, similar applications
Polycarbonate (Lexan)	1,200	2.4	65	130	0.15	Machine windows, guards, impact-resistant shields
Acrylic (PMMA)	1,180	3.2	72	90	0.3	Windows, guards (higher scratch resistance than polycarbonate)
PTFE (Teflon)	2,200	0.5	20-30	260	<0.01	Ultra-low friction bearings, gaskets, seals

Material	Density (kg/m ³)	Young's Modulus (GPa)	Tensile Strength (MPa)	Max Temp (°C)	Water Absorption (%)	Applications
PEEK	1,320	3.6	100	260	0.1	High-performance bearings, seals, electrical insulators
PVC (Rigid)	1,400	3.0	52	60	0.04	Ducting, chip guards, low-cost enclosures

Selection Guidelines:

For Bearing Surfaces: - **Light loads (<50 N), low speed:** UHMW (best wear resistance, self-lubricating) - **Moderate loads (50-500 N), moderate speed:** Acetal/Delrin (higher strength, better dimensional stability) - **High loads (>500 N):** Bronze bushings or linear ball bearings (plastics inadequate)

For Machine Windows/Guards: - **Impact resistance priority:** Polycarbonate (250x stronger than glass, but scratches easily) - **Scratch resistance priority:** Acrylic (harder surface, but 10x weaker than polycarbonate) - **Best compromise:** Polycarbonate with hard coat (MR-10 coating)

For Electrical Insulation: - **General purpose:** Acetal, Nylon (good dielectric strength, low cost) - **High temperature:** PEEK, G-10 fiberglass (withstand >200°C)

A.2.2 Thermoset Plastics

Material	Density (kg/m ³)	Young's Modulus (GPa)	Tensile Strength (MPa)	Max Temp (°C)	Applications
G-10 Fiber-glass (FR4)	1,850	18	310	150	Electrical insulation, PCB substrates, mounting plates
Phenolic (Bakelite)	1,300	5-10	40-60	150	Electrical insulators, vintage machine components
Epoxy Resin (Un-filled)	1,150	3.0	55	120	Adhesives, coatings, composite matrix

A.2.3 Composite Materials

Material	Density (kg/m ³)	Young's Modulus (GPa)	Tensile Strength (MPa)	Cost (\$/kg)	Notes
Epoxy Granite (Poly- mer Con- crete)	2,300-2,500	25-45	50-80	\$2-\$5	Excellent vibra- tion damp- ing (5-10x cast iron), ther- mal stabil- ity. Used for ma- chine beds. DIY- friendly.
Carbon Fiber UD (Unidi- rec- tional)	1,550	135-150	1,500-2,000	\$30-\$80	Highest stiffness- to- weight ratio. Fiber direc- tion critical. Used in aerospace, high- speed gantries.
Carbon Fiber Woven	1,550	70-80	600-800	\$25-\$60	Balanced proper- ties in- plane. Easier to layup than UD. Used for panels, tubes.

Material	Density (kg/m ³)	Young's Modulus (GPa)	Tensile Strength (MPa)	Cost (\$/kg)	Notes
Fiberglass (E-glass/Epoxy)	1,850	25-35	300-500	\$5-\$15	Lower cost than carbon. Good electrical insulation. Used for enclosures, insulators.
Granite (Natural Stone)	2,700	50-70	10-20 (compression)	\$8-\$20	Traditional metrology base material. Excellent thermal stability and damping. Heavy (2.7x denser than aluminum).

Epoxy Granite Formulation Example:

Typical mix ratio (by weight): - Granite aggregate (crushed): 70-80% - Epoxy resin: 15-20% - Hardener: 5-10% - Fillers/additives: 0-5%

Properties: - Young's modulus: 35 GPa (steel = 200 GPa, but epoxy granite has 1/3 density) - Damping ratio: 0.015-0.030 (cast iron: 0.003, steel: 0.0001) - Thermal conductivity: 1.5 W/(mK) (*low -> minimal thermal gradients*) - *Thermal expansion: 12-15 $\mu\text{m}/(\text{m}^\circ\text{C})$ (similar to steel)*

DIY Machine Bed Construction: 1. Build mold from melamine-coated plywood 2. Install inserts for linear rails (threaded bushings in mold) 3. Mix epoxy resin and aggregate (mixer required, 5-10 min mixing) 4. Pour/vibrate to remove air bubbles 5. Cure 24-48 hours at room temperature (or 4-8 hours at 60°C) 6. Machine flat surfaces after cure (grind or scrape)

Advantages over cast iron: - DIY-friendly (no foundry required) - Customizable shape (complex molds possible) - Better damping (reduces chatter in machining) - Corrosion-resistant - Embeds metal inserts easily (rail mounting)

Disadvantages: - Lower stiffness than cast iron (larger cross-sections needed) - Creep over time under sustained load (1-2% over years) - Cannot be welded or brazed

A.3 Material Selection Guidelines by Application

A.3.1 CNC Machine Frame/Base

Requirement: High stiffness, damping, thermal stability, low cost

Material	Stiffness	Damping	Thermal Stability	Cost	Weight	Verdict
Welded Steel Tubing					Heavy	Best for budget builds. Easy to fabricate, weldable.
Cast Iron					Heavy	Best for precision machines. Requires foundry or purchase castings.
Epoxy Granite					Heavy	Best for DIY precision builds. Damping + thermal stability.
Aluminum Extrusion					Light	Best for portable/modular machines. 80/20 ecosystem.

A.3.2 Gantry Beam (Horizontal Spanning Member)

Requirement: High bending stiffness (I), low weight, torsional rigidity

Material / Shape	Stiffness-to-Weight	Deflection (1m span, 1kN center load)	Cost	Verdict
Steel Rectangular Tube (100x50x5mm)	Reference	0.85 mm	\$	Standard choice. Good stiffness, weldable.

Material / Shape	Stiffness-to-Weight	Deflection (1m span, 1kN center load)	Cost	Verdict
Aluminum Rectangular Tube (100x50x5mm)	73% of steel	1.17 mm	\$\$	Lighter but less stiff than steel (same cross-section).
Carbon Fiber Tube (100x50x3mm wall)	180% of steel	0.47 mm	\$\$\$\$	Best stiffness-to-weight. Expensive, requires specialized fabrication.
80/20 Extrusion (15 Series, 3x1.5")	40% of steel tube	2.1 mm	\$\$	Easiest to assemble (modular). Lower stiffness requires larger sections.

Design Rule: Hollow tubes provide best stiffness-to-weight ratio. Torsional rigidity of rectangular tube 3-5x better than I-beam of same weight.

A.3.3 Linear Motion Shafting

Requirement: Hardness for bearing wear, straightness, corrosion resistance

Material	Hardness (HRC)	Straightness	Corrosion Resistance	Cost	Applications
Hardened Chrome-Plated Steel	58-62	+/-0.01 mm/300mm	Good (chrome layer)	\$	Standard linear bearing shafts. Inductive hardened, chrome-plated.
Stainless 440C (Hardened)	58-60	+/-0.01 mm/300mm	Excellent	\$\$	Corrosive environments (waterjet, marine). Full stainless (no plating).

Material	Hardness (HRC)	Straightness	Corrosion Resistance	Cost	Applications
Case-Hardened 1045	50-55 (case)	+/-0.02 mm/300mm	Fair (requires coating)	\$	Budget shafts. Soft core (tough), hard surface (wear-resistant).

Diameter Selection: - 8mm: Light duty, <50 N load - 12mm: Medium duty, 50-200 N load
- 16mm: Heavy duty, 200-500 N load - 20mm+: Very heavy duty, >500 N load

A.3.4 Fasteners (Bolts/Screws)

Requirement: Strength, corrosion resistance, cost

Material / Grade	Tensile Strength (MPa)	Corrosion Resistance	Cost	Applications
Grade 8.8 Steel (Black Oxide)	800	Poor (rust indoors)	\$	Standard structural fasteners. Most common.
Grade 10.9 Steel (Black Oxide)	1,040	Poor	\$	High-stress joints (motor mounts, spindle clamps).
Stainless 304 (A2-70)	700	Excellent	\$\$	Corrosive environments. Slightly lower strength than 8.8 steel.
Stainless 316 (A4-80)	800	Superior (marine grade)	\$\$\$	Waterjet, marine, chemical exposure.

Torque Specifications: See Appendix B.1 for detailed torque tables.

A.4 Material Cost Comparison (USD per kg, Bulk)

Material	Cost (\$/kg)	Relative Cost
Mild Steel (A36)	\$0.50-\$1.00	
Structural Steel Tubing	\$1.00-\$2.00	
Aluminum 6061	\$3.00-\$5.00	
Aluminum 7075	\$8.00-\$15.00	
Stainless 304	\$4.00-\$8.00	
Cast Iron (Castings)	\$2.00-\$5.00	
Epoxy Granite (DIY)	\$2.00-\$5.00	
Carbon Fiber (Prepreg)	\$30.00-\$80.00	
Titanium Ti-6Al-4V	\$25.00-\$50.00	

Small Quantity Markup: Expect 2-5x prices for small quantities (<10 kg) vs. bulk (>100 kg).

A.5 Thermal Properties Summary

Critical for precision machines: Materials with low thermal expansion and high thermal conductivity minimize dimensional changes from temperature fluctuations.

Material	Thermal Expansion ($\mu\text{m}/(\text{m}^{\circ}\text{C})$)	Thermal Conductivity ($\text{W}/(\text{m}^{\circ}\text{K})$)	Thermal Expansion x Length ($\mu\text{m}/^{\circ}\text{C}$ for 1m)
Aluminum 6061	23.6	167	23.6
Steel (Mild)	11.7	50	11.7
Stainless Steel 304	17.3	16	17.3
Cast Iron	10.8	50	10.8
Titanium	8.6	7.2	8.6
Epoxy Granite	12-15	1.5	13.5
Carbon Fiber (axial)	-0.5 to +2	5-10	~0

Example: 1 meter aluminum beam expands 23.6 μm per $^{\circ}\text{C}$ temperature change. 10°C temperature swing = 236 μm (0.236 mm) dimensional change.

Thermal Compensation Strategies: 1. **Material Selection:** Use cast iron or epoxy granite (low expansion) for precision machine bases 2. **Temperature Control:** Enclose machine, use chillers to stabilize temperature ($\pm 1^{\circ}\text{C}$) 3. **Software Compensation:** Measure temperature, apply correction factors in CNC control 4. **Symmetric Design:** Pair materials with matched expansion coefficients

End of Material Properties Appendix